

Categorical + Numeric Problem Set

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DS 5030-001

Q1 [red, blue, blue, red, green, gray]

a)

color	c_1	c_2	c_3
blue	0	0	0
green	1	0	0
gray	0	1	0
red	0	0	1

baseline

b) Sample proportion

$$\hat{p}_{\text{green}} = \frac{\sum I\{c_i = \text{green}\}}{N} = \frac{1}{6}$$

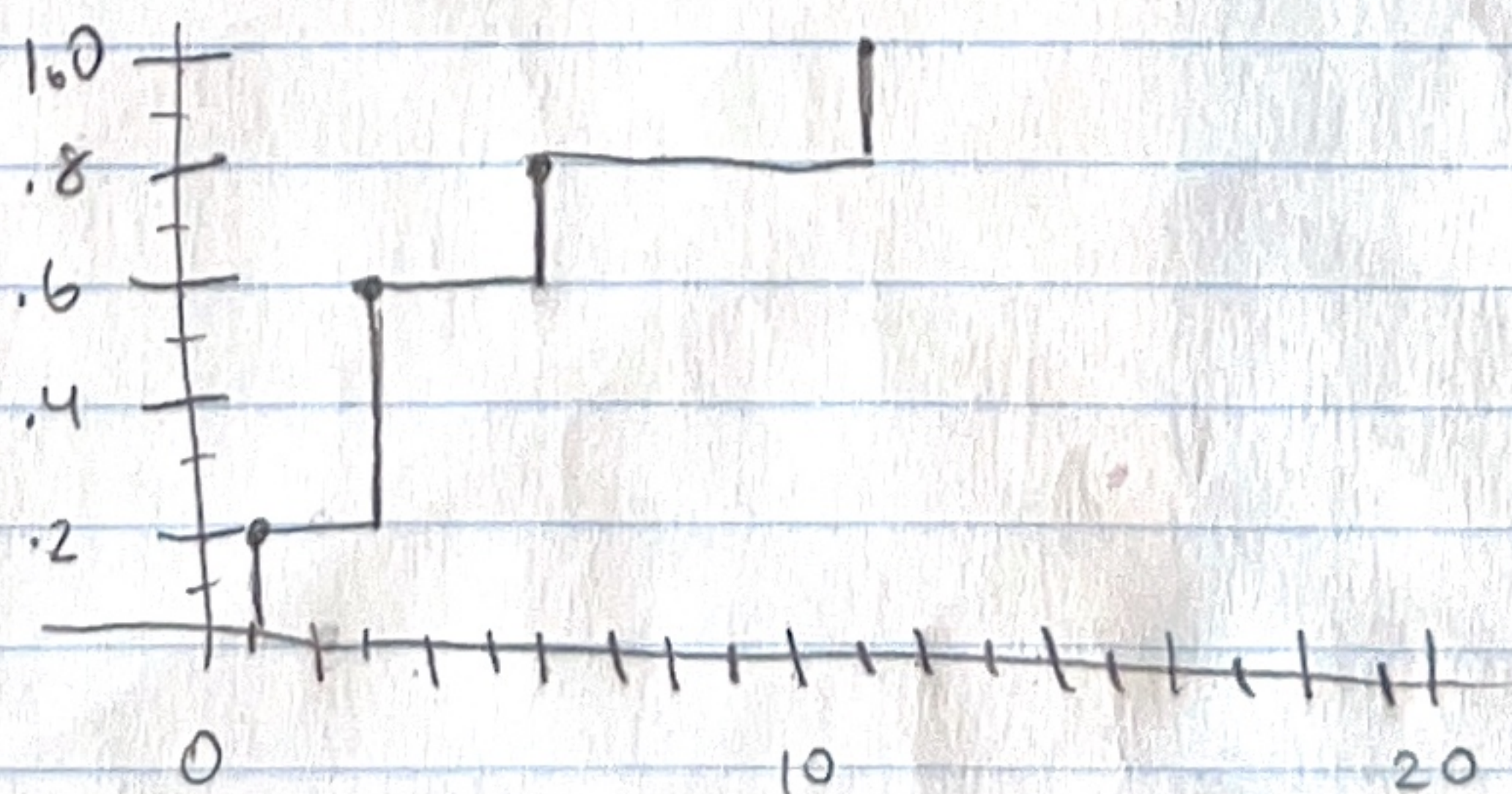
$$\hat{p}_{\text{gray}} = \frac{1}{6}$$

$$\hat{p}_{\text{red}} = \frac{2}{6} = \frac{1}{3}$$

$$\begin{aligned}\hat{p}_{\text{blue}} &= 1 - (\hat{p}_{\text{green}} + \hat{p}_{\text{gray}} + \hat{p}_{\text{red}}) \\ &= 1 - \left(\frac{1}{6} + \frac{1}{6} + \frac{1}{3}\right) \\ &= 1 - \left(\frac{2}{3}\right) \\ &= \frac{1}{3}\end{aligned}$$

$$\hat{p}_i = \begin{cases} \frac{1}{3} & , \text{blue} \\ \frac{1}{6} & , \text{green} \\ \frac{1}{6} & , \text{gray} \\ \frac{1}{3} & , \text{red} \end{cases}$$

c.) $X = [1, 3, 3, 6, 11]$



Q2

a) $Y = a + bX$

$$F_X(x) = P[X \leq x] \quad \text{so} \quad F_Y(y) = P[a + bX \leq y]$$

$$= P[bX \leq y - a]$$

$$= P\left[X \leq \frac{y - a}{b}\right]$$

$$F_Y(y) = \left(\frac{y - a}{b}\right)$$

$$f_Y(y) = \frac{d}{dy} \left(\frac{y - a}{b}\right)$$

$$= \frac{d}{dy} \left(\frac{y}{b} - \frac{a}{b}\right)$$

$$f_Y(y) = 1/b$$

b) $\text{mean}_X = \frac{\sum X_i}{N} = \frac{(1+3+4)}{3} = \left(\frac{8}{3}\right)$

c) $\text{mean}_{X^2} = \frac{\sum X_i^2}{N} = \frac{(1^2 + 3^2 + 4^2)}{3} = \frac{(1 + 9 + 16)}{3}$

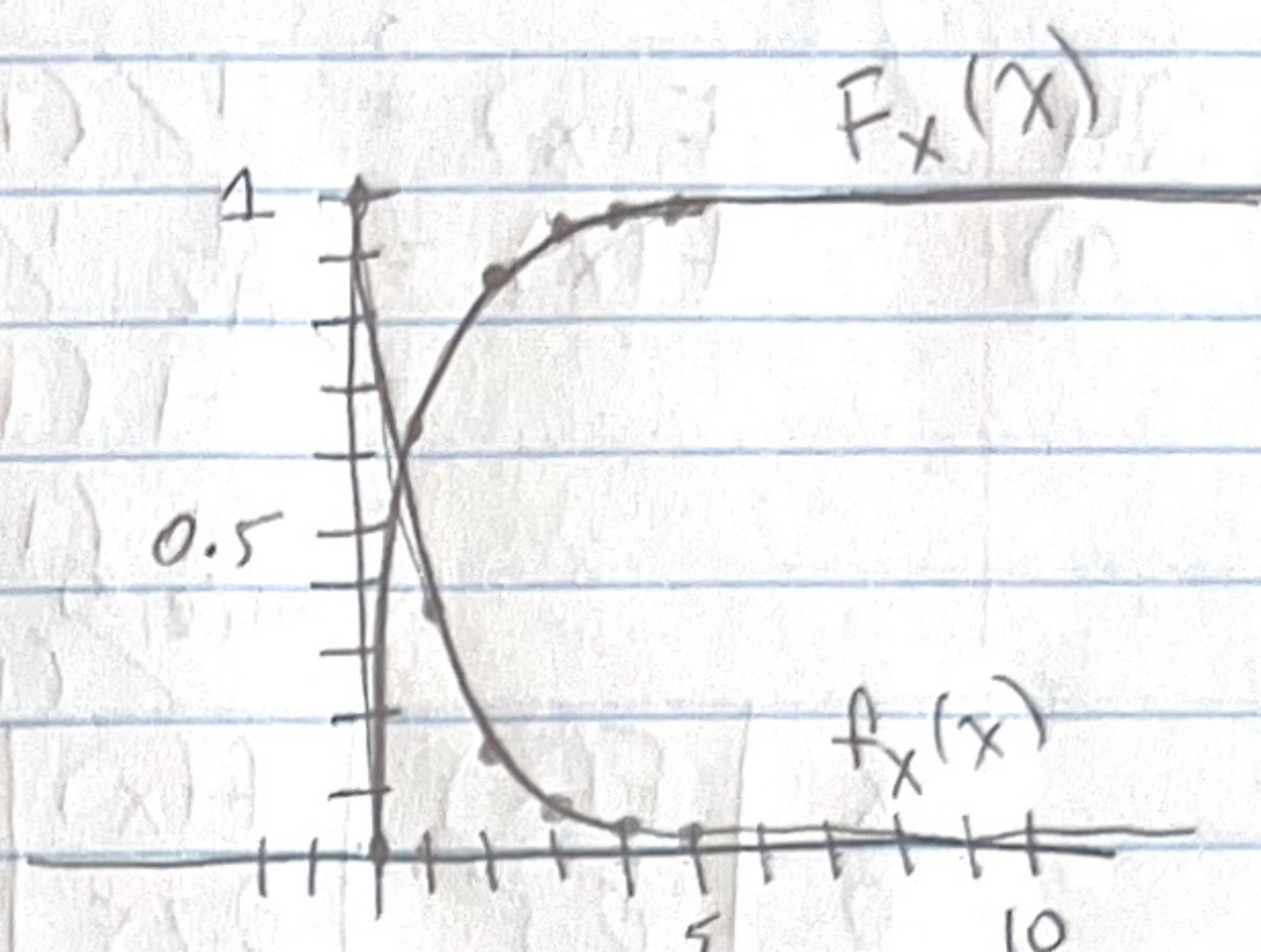
$$= \left(\frac{26}{3}\right)$$

d) $(\text{mean}_X)^2 = \left(\frac{8}{3}\right)^2 = \left(\frac{64}{9}\right)$

No, it does not equal the mean of the square from part c. $\frac{26}{3} \neq \frac{64}{9}$

Q3
a)

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$



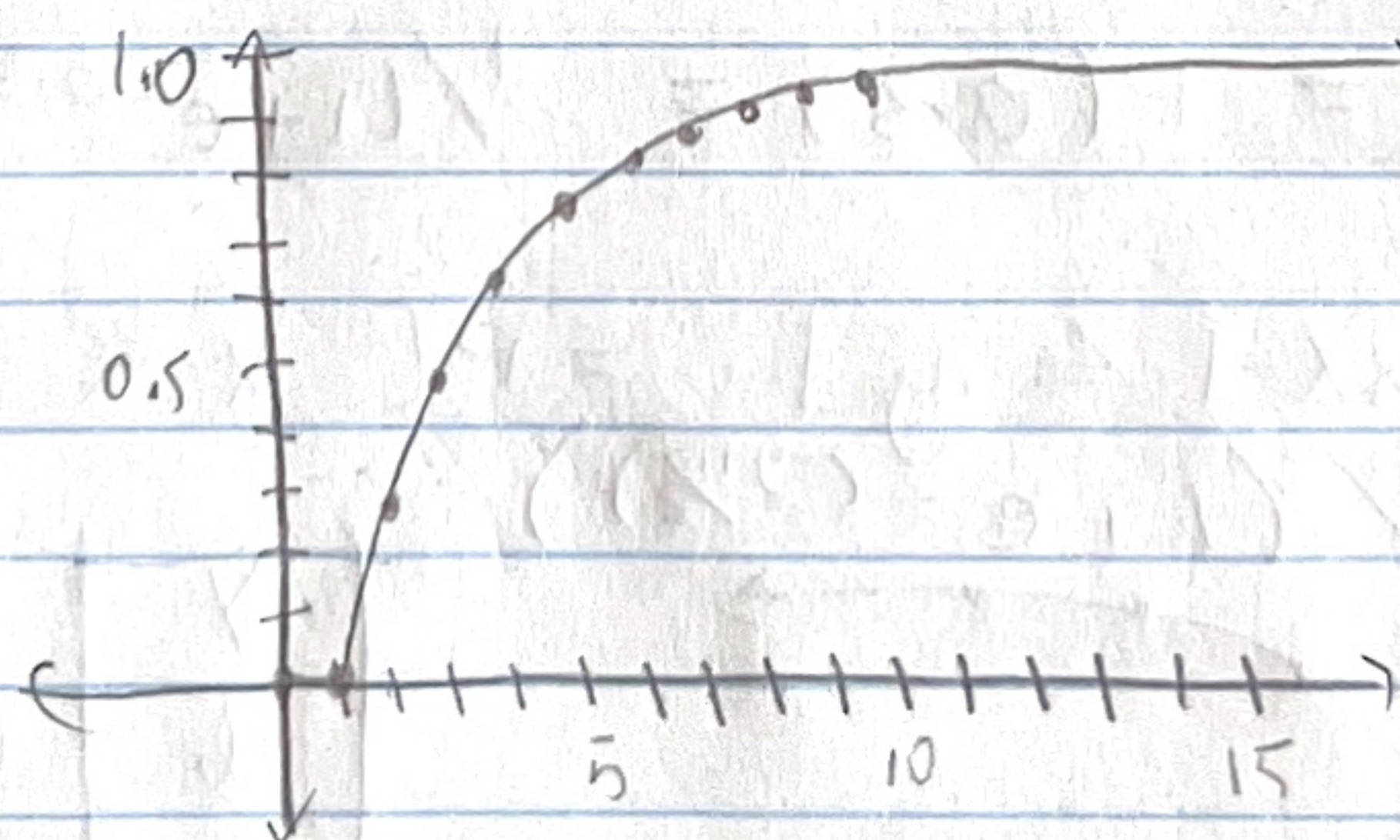
x	F(x)	f(x)
0	0	1
1	0.63	0.37
2	0.87	0.14
3	0.95	0.05
4	0.98	0.02
5	0.99	0.01

$$f(x) = \frac{d}{dx} (1 - e^{-x}) = 0 + e^{-x} = e^{-x}$$

b) $P(X \geq 3) = 1 - P(X < 3) = 1 - 0.95 = 0.05$
 $P(X \leq 35) = 1 - e^{-35} \approx 1$

c) $F_Y(y) = \text{pr}[1 + 3X \leq y] = \text{pr}[3X \leq y - 1]$
 $= \text{pr}[X \leq (y-1)/3] \rightarrow F_Y(y) = (y-1)/3$

$$F_Y(y) = \begin{cases} 0, & y < 1 \\ 1 - e^{-((y-1)/3)}, & y \geq 1 \end{cases}$$



y	F_Y(y)
0	0
1	0
2	0.28
3	0.419
4	0.63
5	0.74
6	0.81
7	0.86
8	0.90
9	0.93
10	0.95

Shifted 1 unit to the right
and slower growth.

$P(Y < 10) = 1 - e^{-((10-1)/3)} = 0.95$

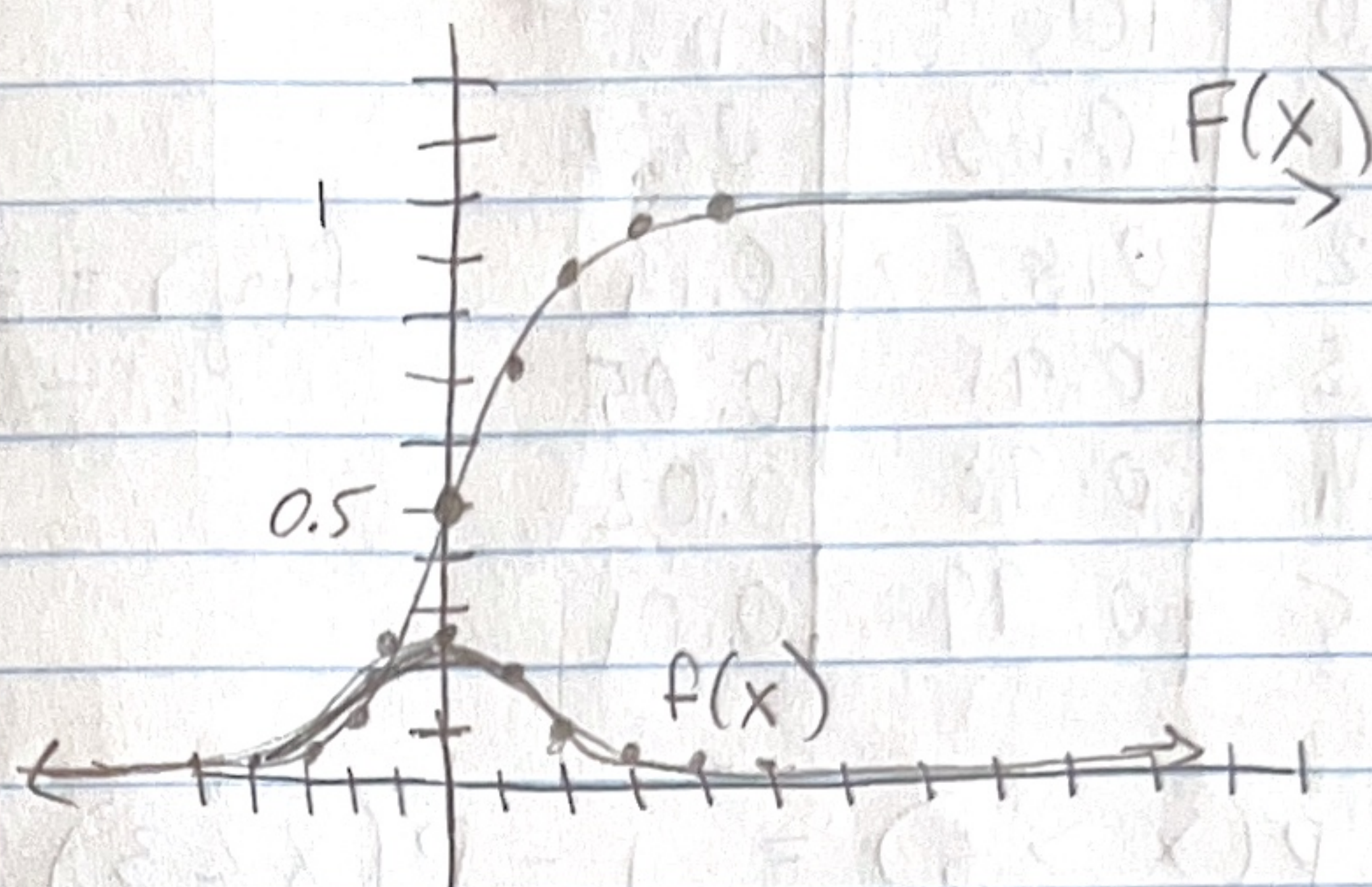
Q4

$$F(x) = 1/(1+e^{-x})$$

a)

$$\begin{aligned} f(x) &= \frac{d}{dx} 1/(1+e^{-x}) = \frac{d}{dx} ((1+e^{-x})^{-1}) \\ &= -1(1+e^{-x})^{-2} * \frac{d}{dx} e^{-x} \\ &= -1(1+e^{-x})^{-2} * -e^{-x} \\ &= \frac{e^{-x}}{(1+e^{-x})^2} \end{aligned}$$

X	F(x)	f(x)
0	0.5	0.25
1	0.73	0.197
2	0.88	0.10
3	0.95	0.05
4	0.98	0.02
5	0.99	0.01
-1	0.26	0.197
-2	0.12	0.10
-3	0.05	0.05
-4	0.02	0.02
-5	0.01	0.01

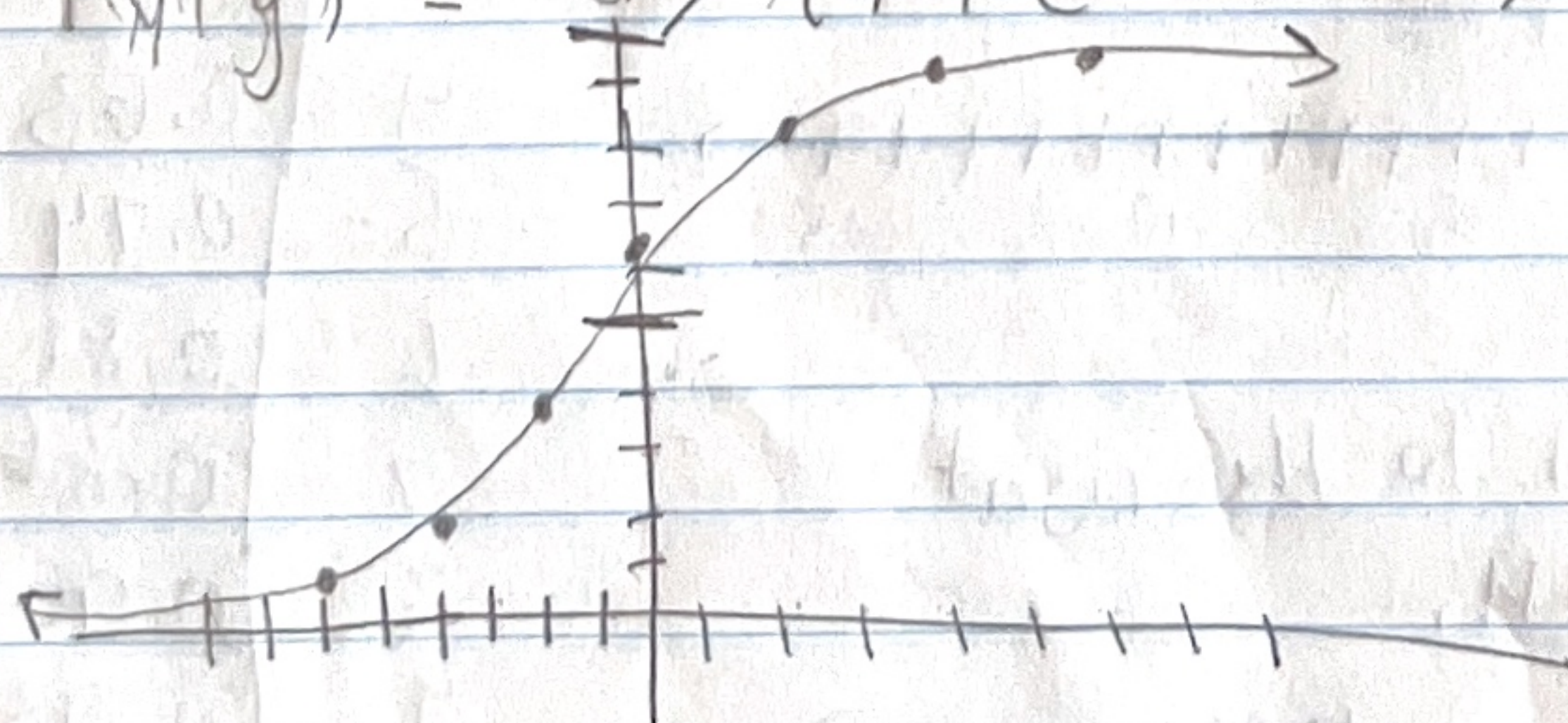


$$\begin{aligned} b) \quad P(X \geq 0.8) &= 1 - F(0.8) = 1 - (1/(1+e^{-0.8})) \\ &= 1 - 0.69 \approx 0.31 \end{aligned}$$

$$P(X \leq 0.3) = F(0.3) = 1/(1+e^{-0.3}) \approx 0.57$$

$$c) \quad F_Y(y) = P[2X-1 \leq y] = P[X \leq (y+1)/2]$$

$$F_Y(y) = 1/(1+e^{-(y+1)/2})$$



$$\begin{aligned} P(Y < 0) &= 1/(1+e^{-(1/2)}) \\ &= 0.622 \end{aligned}$$

y	F(y)
-6	0.08
-4	0.18
-2	0.38
0	0.62
2	0.82
4	0.92
6	0.97

Q5.

b) median of logistic distribution

$$P(X \leq a) = 0.5$$

$$\frac{1}{1+e^{-a}} = 0.5$$

$$\rightarrow 1 = 0.5(1+e^{-a})$$

$$1 = 0.5 + 0.5(e^{-a})$$

$$0.5 = 0.5(e^{-a})$$

$$1 = e^{-a}$$

$$\ln 1 = \ln e^{-a}$$

$$0 = a$$

switched

a) median of exponential distribution

$$1 - e^{-x} = 0.5$$

$$0.5 = e^{-x}$$

$$\ln(0.5) = \ln e^{-x}$$

$$-0.69 = -x$$

$$0.69 = x$$

c) quantile function of exponential distribution

$$cdf = 1 - e^{-x}$$

$$Q_a = 1 - e^{-Q_a}$$

$$e^{-Q_a} = 1 - a$$

$$1/e^{Q_a} = 1 - a$$

$$e^{Q_a} = 1/(1-a)$$

$$\ln(e^{Q_a}) = \ln(1/(1-a))$$

$$Q_a = \ln(1/(1-a)) \text{ where } a \text{ is a quantile in } (0,1)$$

d) quantile function of logistic distribution

$$cdf = 1/(1+e^{-x})$$

$$a = 1/(1+e^{-Q_a})$$

$$1 + e^{-Q_a} = 1/a$$

$$e^{-Q_a} = \frac{1}{a} - 1$$

$$\ln(e^{Q_a}) = a - 1$$

$$Q_a = \ln(a-1) \text{ where } a \text{ is a quantile in } (0,1)$$

also written

$$e^{-Q_a} = \frac{1-a}{a}$$

$$\ln(e^{Q_a}) = \frac{a}{1-a}$$

$$Q_a = \ln(a/(1-a))$$